

logue or mixed-signal design. There is no real penalty, for example, with the inclusion of a 32-bit Cortex-M0 processor core in a design. The core can be implemented utilising a minimum of 12,000 logic gates and so it is virtually negligible in size compared to the surrounding analogue circuitry, which is larger perhaps by a factor of 10:1 and therefore any additional cost for increased die area is correspondingly low.

Also, the addition of a digital subsystem brings a significantly improved ability to test the device—a huge benefit that has not been fully recognised to date. A digital processor becomes a flexible engine that can be used to do all types of on-chip tests and calibration in mixed-signal system-on-chip ICs, enabling much easier validation of the analogue electronics. Another advantage is increased system integration and therefore lower overall cost with the end system. The chances are that a stand-alone microcontroller is already used in a customer design alongside an analogue or mixed-signal chip. An analogue chip vendor is now in a much better position to offer a greater level of integration.

Of course, this may or may not actually happen. Some vendors may wish to use the digital processor for on-chip testability, for example, and not necessarily open it up to customers for further feature development. But overall, this is a very clear value proposition for an analogue chip vendor. And, on the other side of the equation, leading digital vendors are seeking to augment their digital-processing portfolio with analogue components.

Improving energy efficiency of the system

Whilst this new and emerging generation of mixed-signal devices has the capability to fulfil the requirements of the Internet of Things, the big point to make is that embedding a digital processor will significantly improve the efficiency and performance of the overall system, representing huge savings in power consumption.

So what can digital do to enhance the capabilities of analogue components to improve system power consumption? One specific and very important example is in motor control, where power electronics and the analogue are absolutely crucial. A digital processor can enable significantly better motor management of speed and/or rotation, delivering a potential saving in power consumption of around 40 per cent. And the motors in this example are large devices in industrial applications, not necessarily those in low-cost consumable products, such as toys. The digital element does not replace or take anything away from the analogue's side—rather it simply enables digital control to improve the quality of the overall system.

So, with these advanced capabilities, in conjunction with the increased integration possibilities and higher levels of on-chip functionality, the return on investment can be very fast with the inclusion of a small and powerful digital processor core.

The corroboration of all this is that analogue and mixed-signal devices that embed digital-processing capabilities are now coming on to the market. One example is a mixed-signal processor from Analog Devices: the ADSP-CMX40X includes a Cortex-M4 processor and enables motor drive developers to add more features into the silicon.

The device specifically targets energy-efficient applications in industrial markets and is designed to deliver more precise motor control in motor drives, photovoltaic inverters and closed-loop servo control. A second instance is Infineon's XMC4000 family, which also embeds a Cortex-M4 processor; this device is also aimed at improving energy efficiency in industrial applications. And a third is the recent licensing by Dialog Semiconductor of the Cortex-M0 processor, which the company plans to use as the controller in a range of power management ICs for power control and battery management functions in smartphones or tablets.

The future

In a few years' time, or even sooner, it could be that the vast majority, or at least a high percentage, of analogue and mixed-signal chip designs will integrate a high-performance digital processing subsystem to control and manage it, and also enable developers to add their own software code on top.

The key applications are likely to be in motor control and power electronics, and today, the vast majority of these applications are essentially 'dumb.' Endowing them with digital intelligence, in the case of motors, could potentially save enormous amounts of energy. One estimate made recently is that electric motors consume 40 per cent of the world's electrical energy and much of that is in industrial motor drives. So, a little digital treatment of analogue could slash this footprint enabling significant environmental and cost benefits.

In the coming years, as the Internet of Things becomes a larger reality and the demand for reduced power consumption heightens, the model for the next generation of sophisticated analogue and mixed-signal circuits, is likely to combine 'analogue on the edge'—to sense and actuate—with 'digital at the heart'—to control, make decisions and communicate.

A highly integrated and digitally controlled analogue system-on-chip that has the potential to save 40 per cent of the energy consumption of an industrial motor, whether its from an analogue or a digital semiconductor vendor, is an extremely compelling technology that has every reason to be in much demand in the very near future. ●

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